IT3011 - Theory & Practices in Statistical Modelling

Time Series Analysis & Modelling

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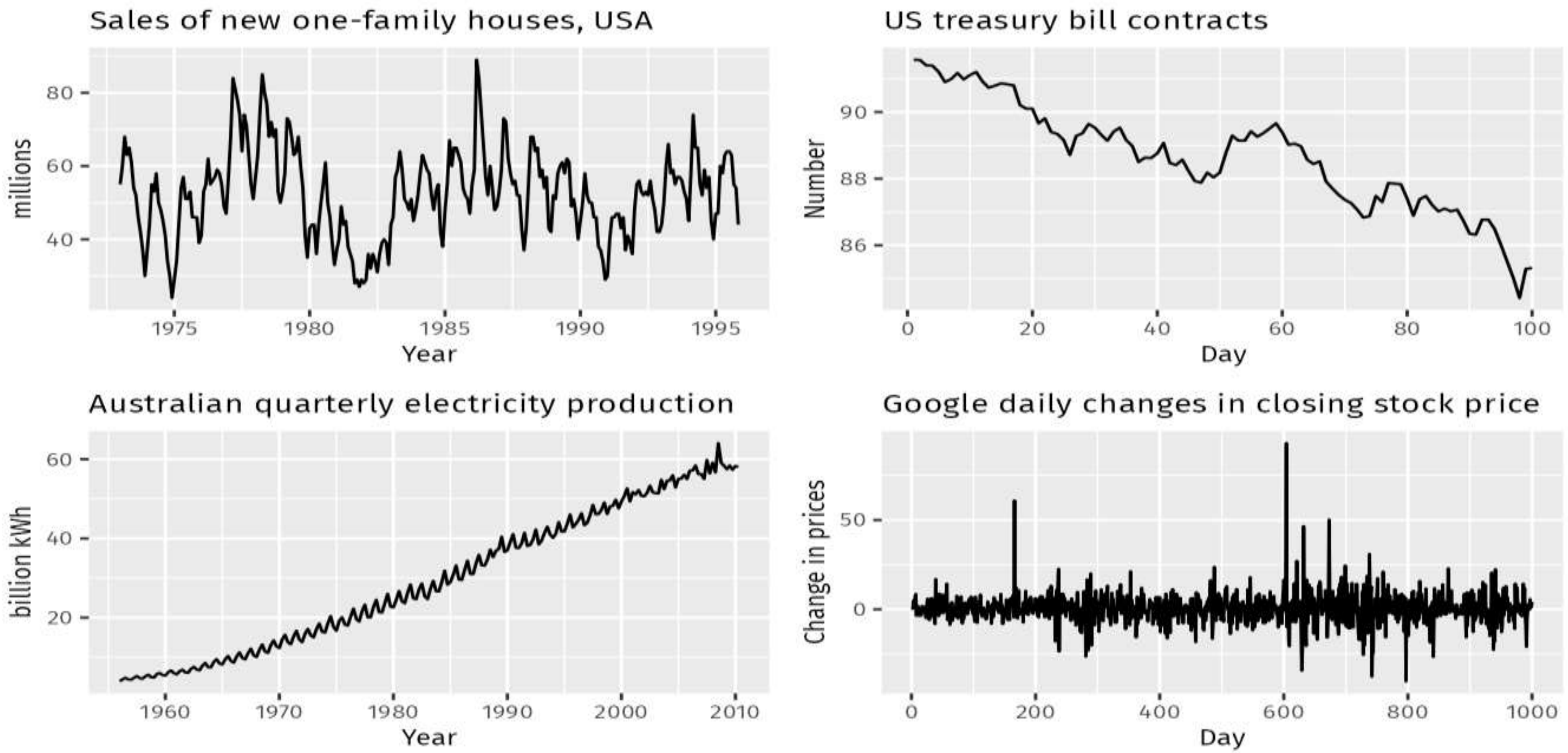
Time Series

- A time series is a data set that tracks a sample over time.
- In particular, a time series allows one to see what factors influence certain variables from period to period.
- Time series analysis can be useful to see how a given asset, security, or economic variable changes over time.

Time Series Patterns

- **Trend** - A trend exists when there is a long-term increase or decrease in the data. It does not have to be linear. Sometimes we will refer to a trend as “changing direction”, when it might go from an increasing trend to a decreasing trend.
- **Seasonal** - A seasonal pattern occurs when a time series is affected by seasonal factors such as the time of the year or the day of the week. Seasonality is always of a fixed and known frequency. The monthly sales of antidiabetic drugs above shows seasonality which is induced partly by the change in the cost of the drugs at the end of the calendar year.
- **Cyclic** - A cycle occurs when the data exhibit rises and falls that are not of a fixed frequency. These fluctuations are usually due to economic conditions, and are often related to the “business cycle”. The duration of these fluctuations is usually at least 2 years.

Time Series Patterns



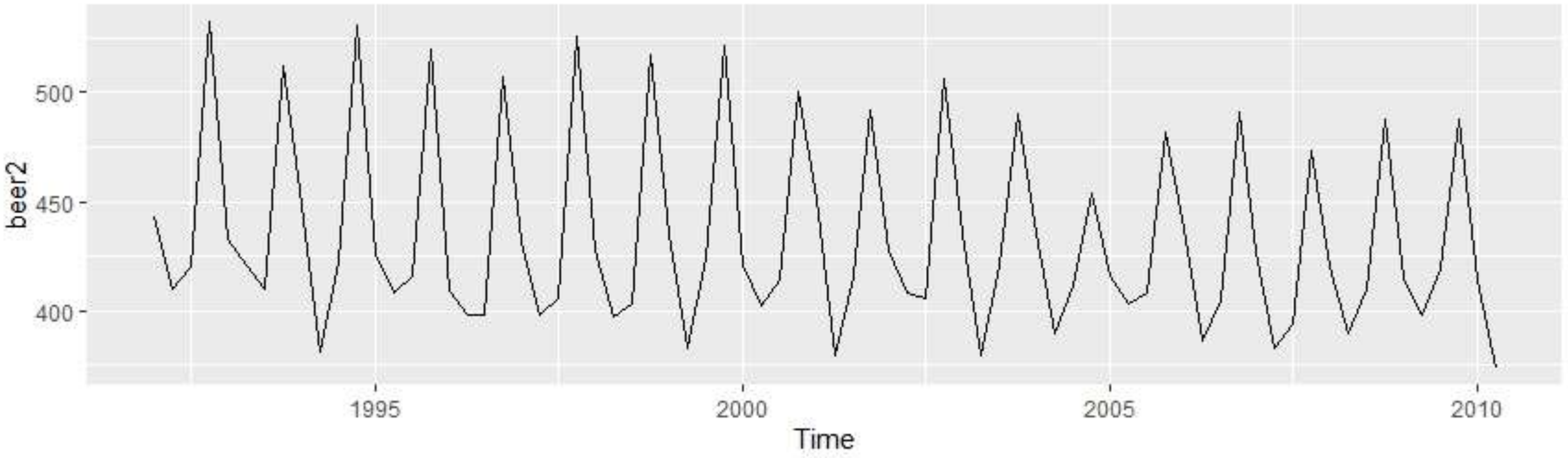
Autocorrelation

- Just as correlation measures the extent of a linear relationship between two variables, autocorrelation measures the linear relationship between lagged values of a time series.
- Where T is the length of the time series.

$$r_k = \frac{\sum_{t=k+1}^{T} (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{T} (y_t - \bar{y})^2}$$

Autocorrelation

- Consider the following time series.



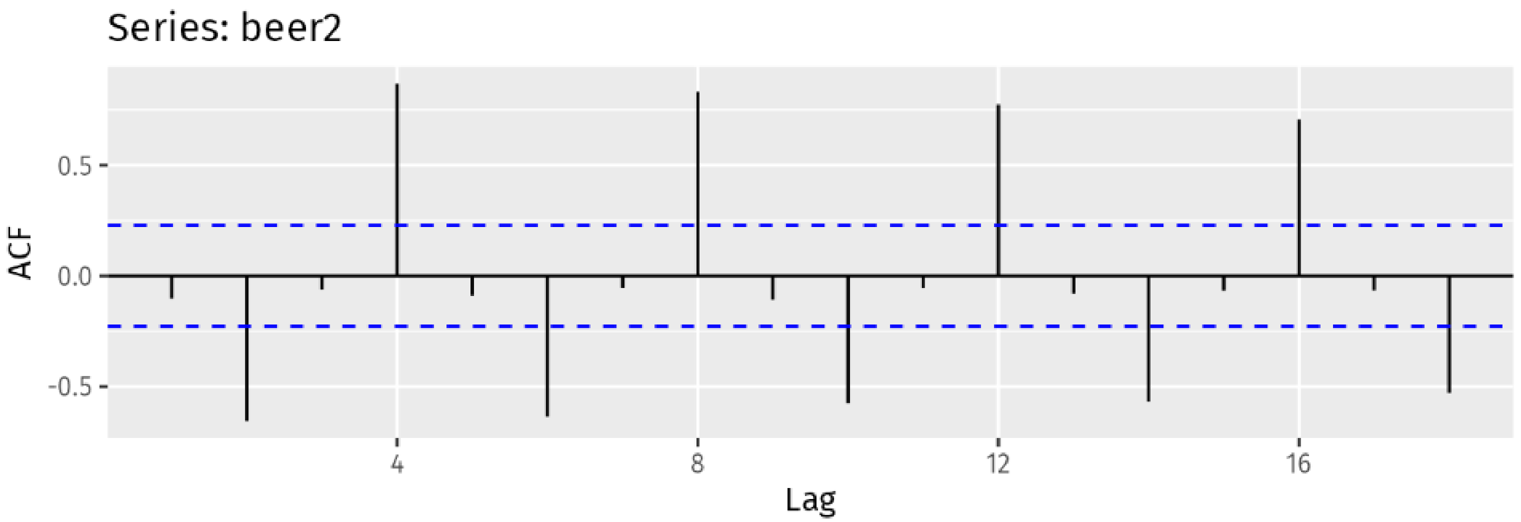
Autocorrelation

- The first nine autocorrelation coefficients for the beer data are given in the following table.

r_1	r_2	r_3	r_4	r_5	r_6	r_7	r_8	r_9
-0.102	-0.657	-0.060	0.869	-0.089	-0.635	-0.054	0.832	-0.108

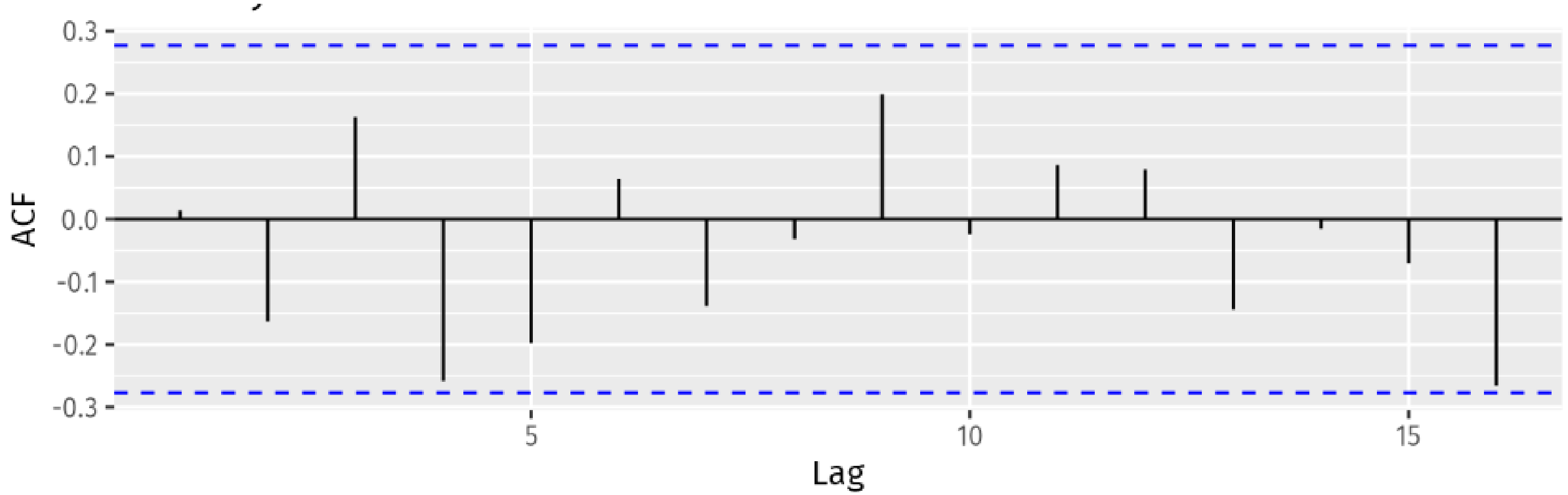
Autocorrelation

- The autocorrelation coefficients are plotted to show the autocorrelation function or ACF.
- The plot is also known as a correlogram.



White Noise

- For white noise series, we expect each autocorrelation to be close to zero. Of course, they will not be exactly equal to zero as there is some random variation.
- For a white noise series, we expect 95% of the spikes in the ACF to lie within $\pm 2/\sqrt{T}$ where T is the length of the time series.



White Noise

- In this example, T=50 and so the bounds are at $\pm 2/\sqrt{50} = \pm 0.28$
- All of the autocorrelation coefficients lie within these limits, confirming that the data are white noise.

Time Series Decomposition

- Time series data can exhibit a variety of patterns, and it is often helpful to split a time series into several components, each representing an underlying pattern category.
- Previously we discussed three types of time series patterns: trend, seasonality and cycles.
- When we decompose a time series into components, we usually combine the trend and cycle into a single trend-cycle component (sometimes called the trend for simplicity).
- Thus, we think of a time series as comprising three components:
    - A trend-cycle component
    - A seasonal component
    - The remainder component (containing anything else in the time series).

Time Series Decomposition

- If we assume an additive decomposition, then we can write,

$$y_t = S_t + T_t + R_t$$

- where y_t is the data, S_t is the seasonal component, T_t is the trend-cycle component, and R_t is the remainder component, all at period t.
- Alternatively, a multiplicative decomposition would be written as,

$$y_t = S_t \times T_t \times R_t$$

Time Series Decomposition

- The additive decomposition is the most appropriate if the magnitude of the seasonal fluctuations, or the variation around the trend-cycle, does not vary with the level of the time series.
- When the variation in the seasonal pattern, or the variation around the trend-cycle, appears to be proportional to the level of the time series, then a multiplicative decomposition is more appropriate. Multiplicative decompositions are common with economic time series.
- An alternative to using a multiplicative decomposition is to first transform the data until the variation in the series appears to be stable over time, then use an additive decomposition.

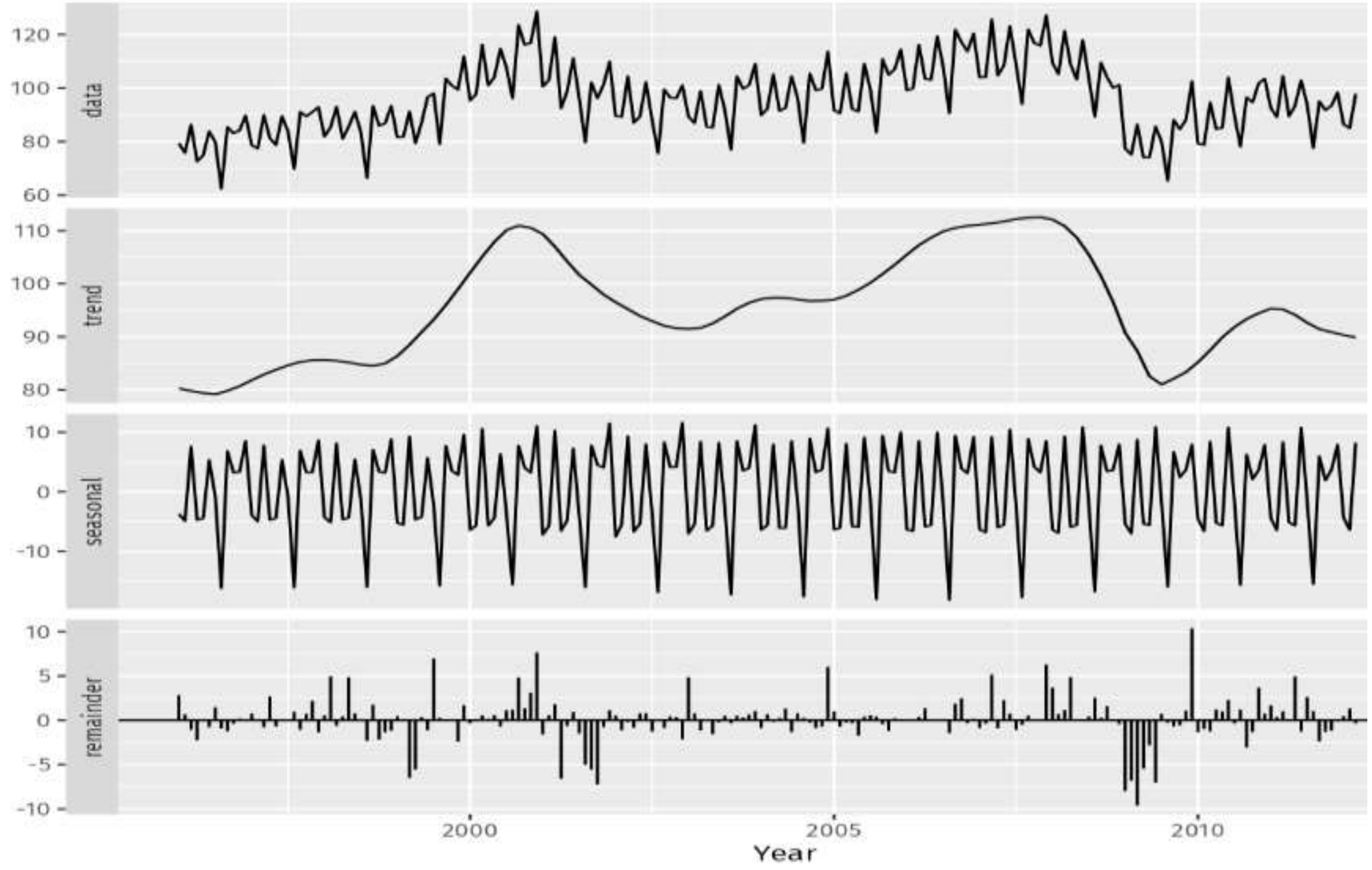
Time Series Decomposition

- When a log transformation has been used, this is equivalent to using a multiplicative decomposition because,

$$y_t = S_t \times T_t \times R_t \quad \text{is equivalent to} \quad \log y_t = \log S_t + \log T_t + \log R_t$$

Time Series Decomposition

- Following figure shows an additive decomposition of these data.



Time Series Decomposition

- The three components are shown separately in the bottom three panels of the figure.
- These components can be added together to reconstruct the data shown in the top panel.
- Notice that the seasonal component changes slowly over time, so that any two consecutive years have similar patterns, but years far apart may have different seasonal patterns.
- The remainder component shown in the bottom panel is what is left over when the seasonal and trend-cycle components have been subtracted from the data.

Classical Decomposition

- The classical decomposition method originated in the 1920s.
- It is a relatively simple procedure, and forms the starting point for most other methods of time series decomposition.
- There are two forms of classical decomposition:
  - Additive decomposition
  - Multiplicative decomposition.
- The first step in a classical decomposition is to use a moving average method to estimate the trend-cycle

Moving Average Smoothing

- A moving average of order **m** can be written as,

$$\hat{T}_t = \frac{1}{m} \sum_{j=-k}^{k} y_{t+j}$$

- where m=2k+1
- That is, the estimate of the trend-cycle at time t is obtained by averaging values of the time series within k periods of t.
- Observations that are nearby in time are also likely to be close in value.
- Therefore, the average eliminates some of the randomness in the data, leaving a smooth trend-cycle component.
- We call this an m-MA, meaning a moving average of order m.

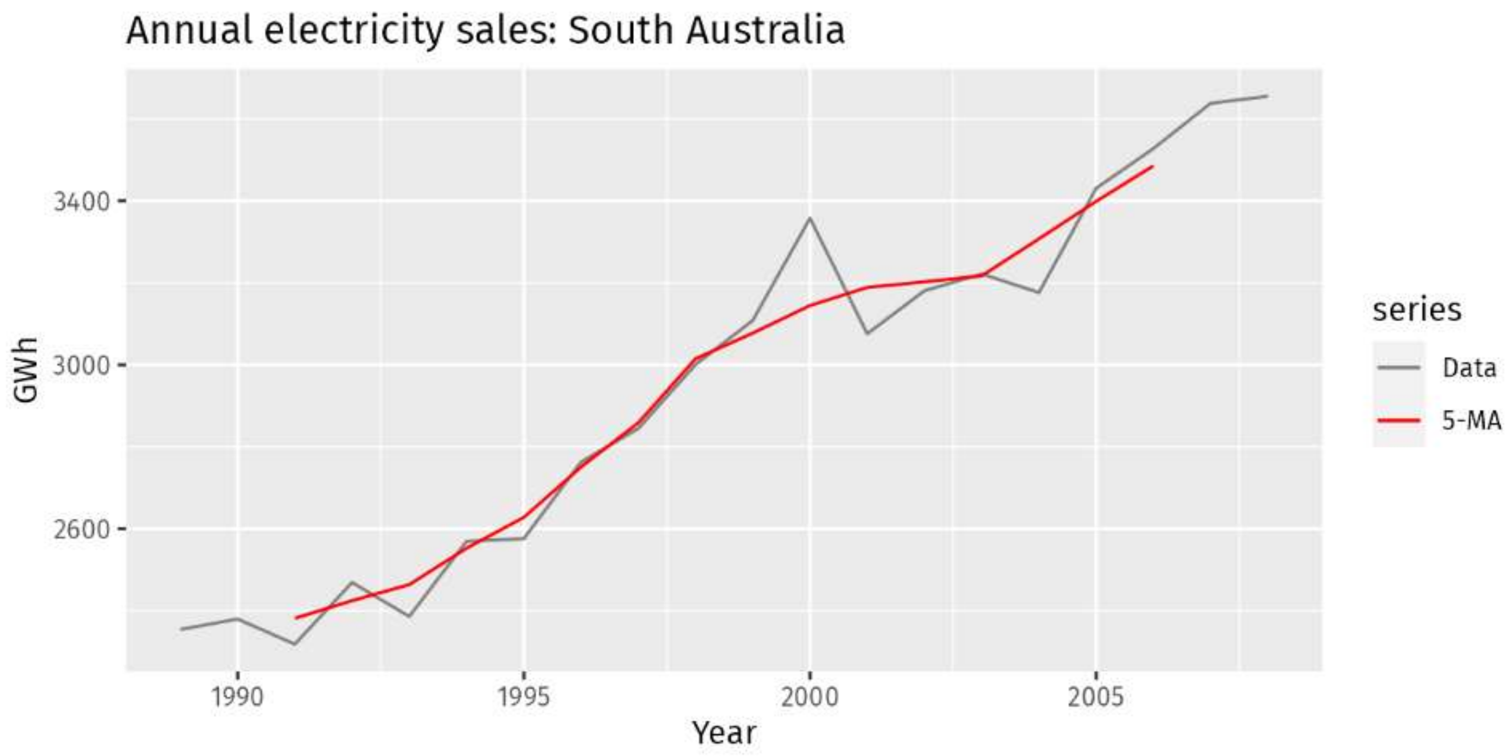
Moving Average Smoothing

- Consider the following example

Year	Sales (GWh)	5-MA
1989	2354.34	
1990	2379.71	
1991	2318.52	2381.53
1992	2468.99	2424.56
1993	2386.09	2463.76
1994	2569.47	2552.60
1995	2575.72	2627.70
1996	2762.72	2750.62
1997	2844.50	2858.35
1998	3000.70	3014.70
1999	3108.10	3077.30
2000	3357.50	3144.52
2001	3075.70	3188.70
2002	3180.60	3202.32
2003	3221.60	3216.94
2004	3176.20	3307.30
2005	3430.60	3398.75
2006	3527.48	3485.43
2007	3637.89	
2008	3655.00	

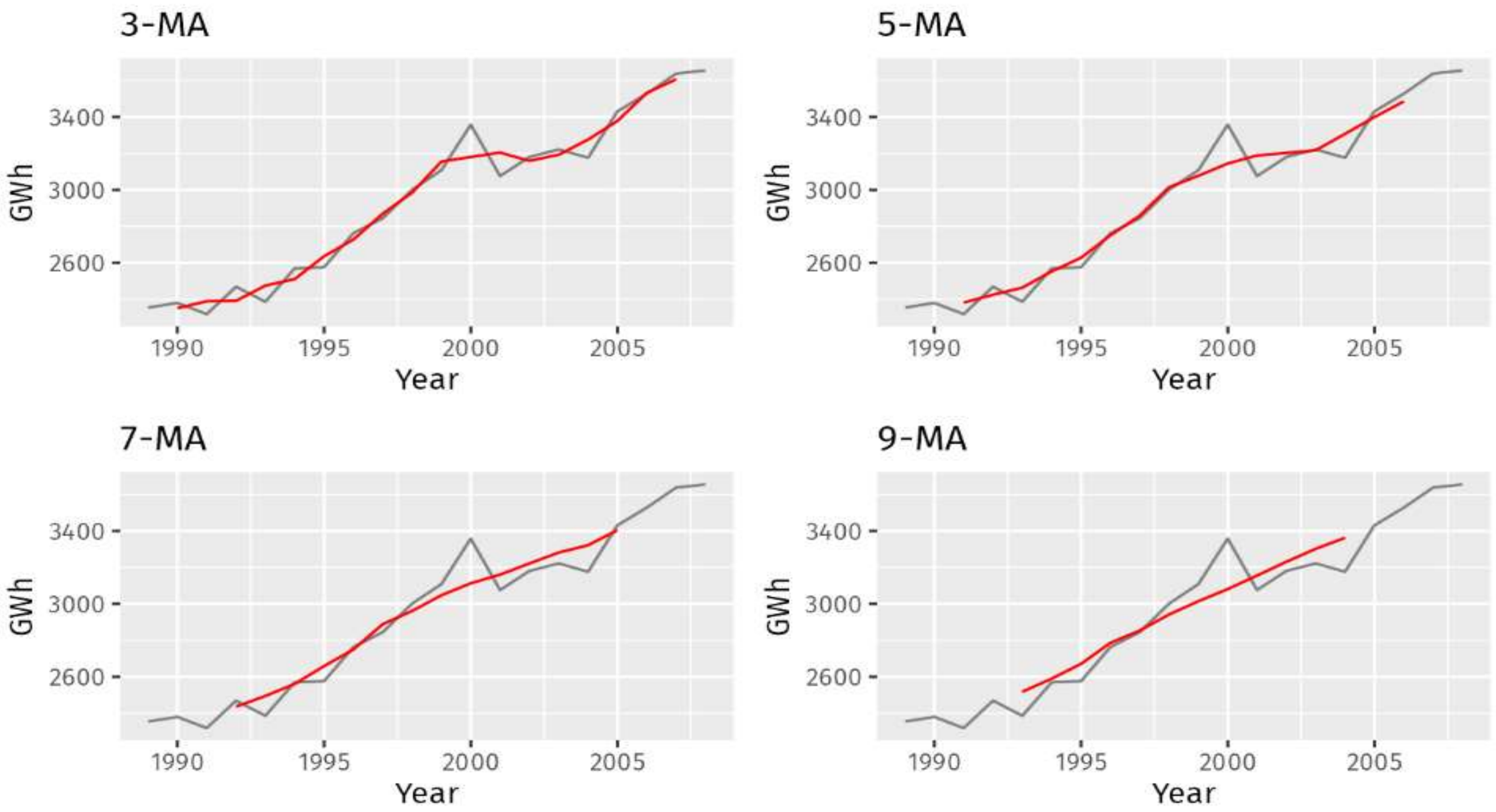
Moving Average Smoothing

- After smoothing



Moving Average Smoothing

- Different Moving Average Smoothing



Additive Decomposition

Step 1

If m is an even number, compute the trend-cycle component $\hat{T}_t$ using a $2 \times m$-MA. If m is an odd number, compute the trend-cycle component $\hat{T}_t$ using an m-MA.

Step 2

Calculate the detrended series: $y_t - \hat{T}_t$.

Step 3

To estimate the seasonal component for each season, simply average the detrended values for that season. For example, with monthly data, the seasonal component for March is the average of all the detrended March values in the data. These seasonal component values are then adjusted to ensure that they add to zero. The seasonal component is obtained by stringing together these monthly values, and then replicating the sequence for each year of data. This gives $\hat{S}_t$.

Step 4

The remainder component is calculated by subtracting the estimated seasonal and trend-cycle components: $\hat{R}_t = y_t - \hat{T}_t - \hat{S}_t$.

Multiplicative Decomposition

Step 1

If m is an even number, compute the trend-cycle component $\hat{T}_t$ using a $2 \times m$-MA. If m is an odd number, compute the trend-cycle component $\hat{T}_t$ using an m-MA.

Step 2

Calculate the detrended series: $y_t/\hat{T}_t$.

Step 3

To estimate the seasonal component for each season, simply average the detrended values for that season. For example, with monthly data, the seasonal index for March is the average of all the detrended March values in the data. These seasonal indexes are then adjusted to ensure that they add to m. The seasonal component is obtained by stringing together these monthly indexes, and then replicating the sequence for each year of data. This gives $\hat{S}_t$.

Step 4

The remainder component is calculated by dividing out the estimated seasonal and trend-cycle components: $\hat{R}_t = y_t/(\hat{T}_t\hat{S}_t)$.

Stationarity

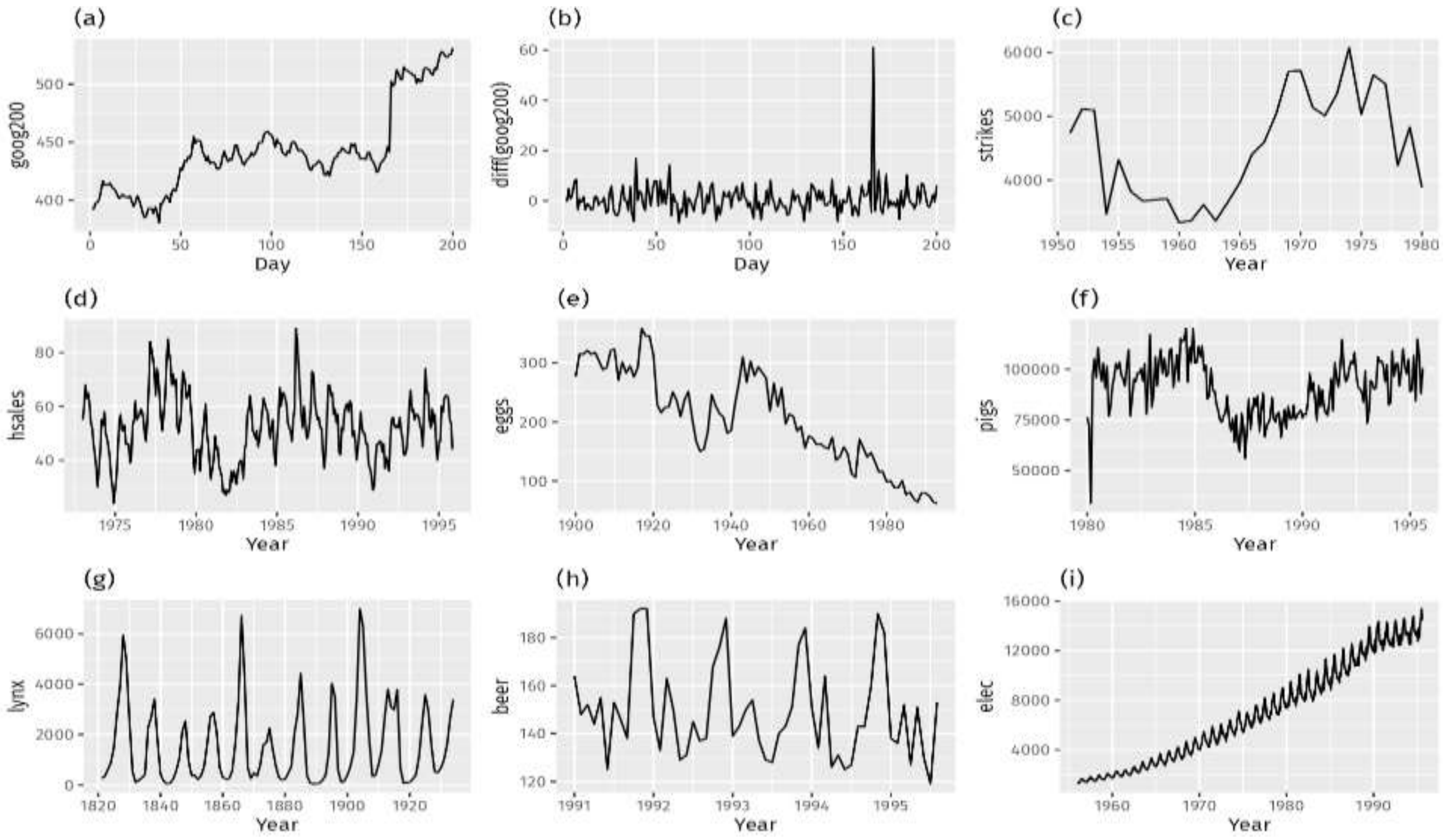
- A stationary time series is a time series where the mean, variance, and autocovariance of the data do not depend on the specific time at which the observations are made.
- Stationarity is an important concept in time series analysis because many time series models and statistical techniques assume or require stationarity for accurate results.
- Stationary time series are easier to analyze and predict because their statistical properties remain consistent over time, allowing patterns and relationships to be more easily identified.

Stationarity

- A stationary time series is one whose properties do not depend on the time at which the series is observed.
- Thus, time series with trends, or with seasonality, are not stationary because the trend and seasonality will affect the value of the time series at different times.
- On the other hand, a white noise series is stationary.
- In general, a stationary time series will have no predictable patterns in the long-term.

Stationarity

- Which of these do you think are stationary?



Stationarity

- Obvious seasonality rules out series (d), (h) and (i).
- Trends and changing levels rules out series (a), (c), (e), (f) and (i). Increasing variance also rules out (i).
- That leaves only (b) and (g) as stationary series.
- At first glance, the strong cycles in series (g) might appear to make it non-stationary.
- But these cycles are aperiodic — they are caused when the lynx population becomes too large for the available feed, so that they stop breeding and the population falls to low numbers, then the regeneration of their food sources allows the population to grow again, and so on.
- In the long-term, the timing of these cycles is not predictable.
- Hence the series is stationary.

Stationarity

- Stationarity can be tested using Augmented Dickey Fuller (ADF) test.

$$H_0: Time - series\ is\ non - stationary$$
$$H_1: Time - series\ is\ stationary$$

Differencing

- This shows one way to make a non-stationary time series stationary — compute the differences between consecutive observations.
- This is known as differencing.
- Transformations such as logarithms can help to stabilize the variance of a time series.
- Differencing can help stabilize the mean of a time series by removing changes in the level of a time series, and therefore eliminating (or reducing) trend and seasonality.

Differencing

- The ACF plot is also useful for identifying non-stationary time series.
- For a stationary time series, the ACF will drop to zero relatively quickly, while the ACF of non-stationary data decreases slowly.
- Also, for non-stationary data, the value of r1 is often large and positive.

Random-Walk Model

- The differenced series is the change between consecutive observations in the original series, and can be written as,

$$y'_t = y_t - y_{t-1}$$

- The differenced series will have only T−1 values.
- When the differenced series is white noise, the model for the original series can be written as,

$$y_t - y_{t-1} = \varepsilon_t$$

- where ε_t denotes white noise. Rearranging this leads to the “random walk” model.

$$y_t = y_{t-1} + \varepsilon_t$$

Random-Walk Model

- Random walk models are widely used for non-stationary data, particularly financial and economic data.
- Random walks typically have:
  - long periods of apparent trends up or down
  - sudden and unpredictable changes in direction.
  - The forecasts from a random walk model are equal to the last observation, as future movements are unpredictable, and are equally likely to be up or down.

Second Order Differencing

- Occasionally the differenced data will not appear to be stationary and it may be necessary to difference the data a second time to obtain a stationary series:

$$\begin{aligned} y''_t &= y'_t - y'_{t-1} \\ &= (y_t - y_{t-1}) - (y_{t-1} - y_{t-2}) \\ &= y_t - 2y_{t-1} + y_{t-2}. \end{aligned}$$

- The differenced series will have only T−2 values.
- Then, we would model the “change in the changes” of the original data.
- In practice, it is almost never necessary to go beyond second-order differences.

Seasonal Differencing

- A seasonal difference is the difference between an observation and the previous observation from the same season.
- So,

$$y'_t = y_t - y_{t-m}$$

- where m=the number of seasons.
- These are also called “lag-m differences”, as we subtract the observation after a lag of m periods.
- If seasonally differenced data appear to be white noise, then an appropriate model for the original data is,

$$y_t = y_{t-m} + \varepsilon_t$$

Backshift Notation

- The backward shift operator **B** is a useful notational device when working with time series lags:

$$By_t = y_{t-1}$$

$$B(By_t) = B^2 y_t = y_{t-2}$$

- This can be written as,

$$B^i y_t = y_{t-i}$$

- Or this can also be written as,

$$y_t B^i = y_{t-i}$$

Backshift Notation

- The backward shift operator is convenient for describing the process of differencing. A first difference can be written as,

$$y'_t = y_t - y_{t-1} = y_t - By_t = (1 - B)y_t$$

- Note that a first difference is represented by (1−B).
- Similarly, if second-order differences have to be computed, then:

$$y''_t = y_t - 2y_{t-1} + y_{t-2} = (1 - 2B + B^2)y_t = (1 - B)^2 y_t$$

Backshift Notation

- In general, a d th-order difference can be written as,

$$(1 - B)^d y_t$$

Time Series Modeling

- Time series modeling involves analyzing and forecasting data points collected over time to understand and predict patterns, trends, and relationships within the series.
- It is widely used in various fields such as economics, finance, weather forecasting, sales forecasting, and more.

Autoregressive (AR) Models

- In a multiple regression model, we forecast the variable of interest using a linear combination of predictors.
- In an autoregression model, we forecast the variable of interest using a linear combination of past values of the variable.
- The term autoregression indicates that it is a regression of the variable against itself.
- Thus, an autoregressive model of order p can be written as,

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t$$

- This is called as **AR(p)** model.

Autoregressive (AR) Models

- The assumptions of autoregressive models include:
- **Stationarity**: Autoregressive models assume that the time series being modeled is stationary or can be transformed into a stationary series. Stationarity implies that the mean, variance, and autocovariance structure of the data remain constant over time.
- **Linearity**: Autoregressive models assume a linear relationship between the current observation and the past observations. This means that the impact of past observations on the current value is assumed to be linear.
- **Independence**: Autoregressive models assume that the residuals (the differences between the observed values and the predicted values) are independent of each other. The residuals should not exhibit any patterns or correlations that can be exploited to improve the model.
- **Homoscedasticity**: Autoregressive models assume that the variance of the residuals is constant over time. In other words, the spread of the residuals does not change as the time series progresses.

Autoregressive (AR) Models

- The assumptions of autoregressive models include:
- **Normality**: Autoregressive models often assume that the residuals follow a normal distribution. This assumption allows for the use of statistical tests and confidence intervals based on normality.

Autoregressive (AR) Models

- Why Stationarity?

- AR models assume stationarity in the time series data.
- Stationarity means that the statistical properties of the data, such as the mean and variance, remain constant over time.
- If the data is non-stationary, it can exhibit trends, seasonality, or other changing patterns that violate the assumptions of the AR model.
- Using non-stationary data in an AR model can lead to unreliable parameter estimates and inaccurate predictions.

Moving Average (MA) Models

- Rather than using past values of the forecast variable in a regression, a moving average model uses past forecast errors in a regression-like model.

$$y_t = \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

- This is called as **MA(q)** model.

ARMA Models

- If we autoregression and a moving average model, we obtain a non-seasonal ARMA model.
- ARMA is an acronym for Auto Regressive Moving Average.
- The full model can be written as,

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

- This is called as **ARMA(p,q)** model.

ARIMA Models

- If we combine differencing with autoregression and a moving average model, we obtain a non-seasonal ARIMA model.
- ARIMA is an acronym for AutoRegressive Integrated Moving Average (in this context, “integration” is the reverse of differencing).
- The full model can be written as,

$$y'_t = c + \phi_1 y'_{t-1} + \cdots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

- This is called as **ARIMA(p,d,q)** model.